

Tianyi ZHANG

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Work Experience

Advanced Micro Devices, Inc. (AMD)	Jul 2024 – Present
Assistant Project Engineer, MSIP Department, SRDC	Shanghai
• IP Development and Verification: Contributed to memory read/write IP development for AMD accelerator projects, supporting RTL design and verification. • SoC Verification Automation: Developed Python and Shell tools for SoC signal checks, traffic analysis, and RTL error injection, using VCS, Verdi, and Perforce to support regression testing and verification analysis. • AI-Assisted RTL Debugging: Developed a skill-selection mechanism for Verilog debugging, scoring specialised skills by error characteristics, debugging stage and design context to select methods for specific RTL issues. • Simulation Feedback and Evaluation: Reran simulations after AI-assisted debugging and reviewed errors and results in simulation logs; evaluated selected skills to guide subsequent RTL debugging iterations. • AI Usage Analytics: Contributed to a real-time code-commit analytics platform that automatically analyses commits and measures AI-assisted development usage, helping engineering teams track AI tool adoption.	

Education

Nanyang Technological University	Aug 2026 – Expected Jun 2027
MSc in Precision Scientific Instrumentation	Singapore
Recipient of an NTU postgraduate scholarship; focus: semiconductor instrumentation, precision electronics, and sensing systems.	
University College London (UCL)	Sept 2023 – Jun 2026
BEng in Mechanical Engineering First Class Honours	London, United Kingdom
Technical Skills	

Digital Verification & EDA: Verilog, Synopsys VCS, Verdi, SoC signal checks, regression testing, RTL error injection
Programming & Tools: Python, C++, Shell, Java, MATLAB, Linux, Git, Perforce
Agentic Coding & ML: Agent Skills, workflow orchestration, evaluation harnesses, scikit-learn, pandas

Projects

Agent Skill: Engineering Automation Workflows and Evaluation	2026
Python / LLM Agents / Workflow Runner / Evaluation Harness	
• Developed modular Agent Skills and a Python workflow runner for engineering automation, with prompt preflight checks, model calls and saved outputs; linked tasks, skills, models and prompts to runtime, token usage and failure logs. • Built an offline evaluation harness with rule-based scoring, batch runs and benchmark comparisons; checked dataset, split and scoring consistency, exported CSV results and regression-tested evaluation logic.	
Model Training: Bank Term Deposit Classification	2026
Python / scikit-learn / imbalanced-learn / pandas	
• Constructed model-training pipelines for 45,211 bank records, comparing logistic regression, SVM, KNN, tree and boosting models, voting and stacking ensembles, and class-imbalance treatments. • Implemented five-fold stratified evaluation using MCC, with preprocessing, sampling and feature selection fitted within training folds to prevent data leakage; excluded post-call duration and logged out-of-fold predictions and training time, and used hashes for run traceability and reproducibility.	
Scooter Modelling: Energy Simulation and Route Optimisation	2026
Python / Graph Algorithms / OpenStreetMap	
• Built a Python simulation converting maps, elevation and traffic into energy-weighted road networks; combined graph search, physical energy modelling and regenerative braking to compare urban routes by energy consumption.	

Campus Activities & Awards

UCL Racing Formula E Team (2023–2026) — Assisted with debugging motor signals and sensor feedback systems.
Vice President & Project Control Lead, Robotics Club (2020–2023) — Organised the assembly and debugging of robotic control modules and participated in the FRC China Regional Competition.
British Physics Olympiad Gold Award (2022).

Languages

Mandarin Chinese (Native); English (IELTS 7.5).